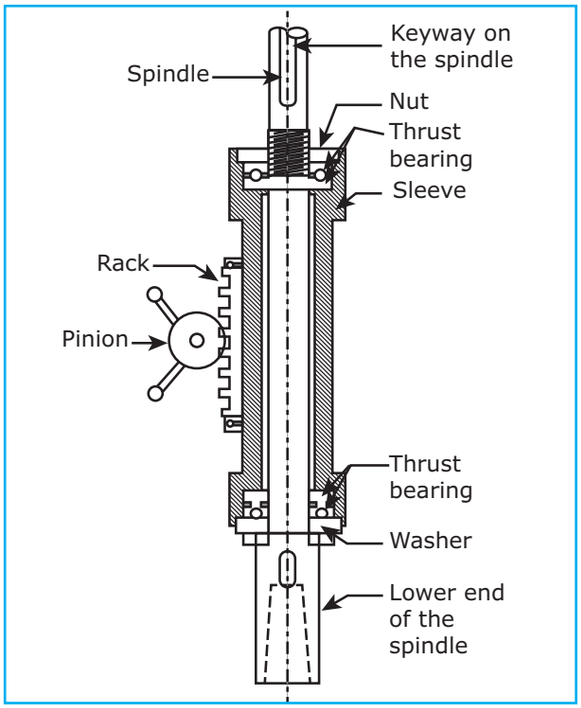


Rack teeth are cut on the outer surface of the sleeve. The sleeve may be moved up and down by rotating a pinion which meshes with the rack. This movement is given to the spindle for providing the required feed. As there is a long keyway on top of the spindle, it is connected to the driving mechanism even during the feed movement.

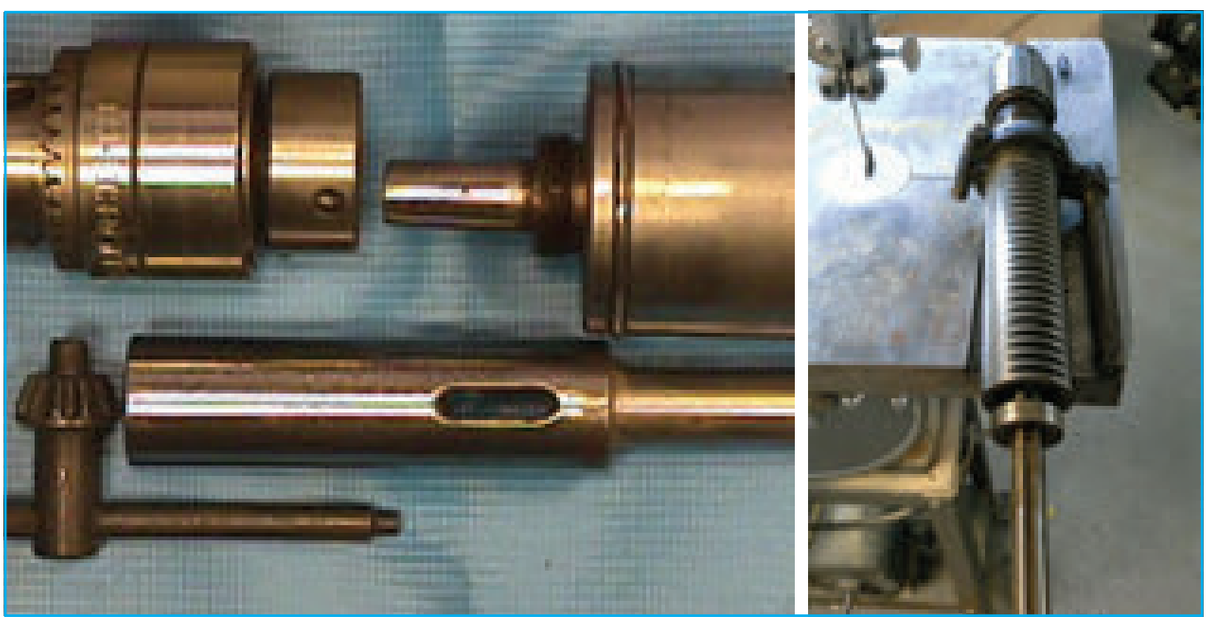
A morse taper hole is provided at the lower end of the spindle. It is useful in accommodating a taper shank drill. The tang of the drill fits into a slot provided at the end of the taper hole. To remove the drill from the spindle a drift may be pushed through the slot.

The spindle drive is obtained in three methods. They are

1. Step cone pulley drive
2. Step cone pulley with back gear arrangement.
3. Gear box drive.

2.14 Work Holding Devices

The work should be held firmly on the machine table before performing any operations on it. As the drill exerts very high quantity of torque while rotating the work should not be held by hand. If the work pieces are not held by a proper holding device, it will start rotating along with the tool causing injuries to the operator and damage to the machine.



Working Principle of the Spindle

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